ECOCHECK

A Community-Driven Mobile App for Recycling and Waste Management Reporting

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ABSTRACT

Waste volume and improper disposal present a tough social concern. In the Philippines, escalating waste volume compounds inadequate waste infrastructure. This situation requires integrated strategies for both refining disposal and easing sustainable consumption. This study developed EcoCheck, a community-driven mobile app for recycling and waste management reporting. It allows residents to report illegal dumping using geo-tagged photos and access proper recycling information. Local Government Units (LGUs) gain actionable data to inform waste infrastructure decisions. This approach uses "citizen science" to enhance environmental literacy and advocacy. The study used a quantitative research design. Researchers evaluated the app using the ISO/IEC 25010 software quality standard. Participants included residents, barangay officials, and IT experts. EcoCheck successfully met its objectives. It provides a practical, scalable model to increase public awareness and supply LGUs with real-time, geo-specific data. Future work should scale implementation and add reward systems to further increase participation. EcoCheck is a feasible, high-quality digital intervention for waste management.

OBJECTIVES

User-Centric Design and Reliability

Assesses the intuitive interface and technical features to ensure effortless reporting and seamless navigation for all residents.

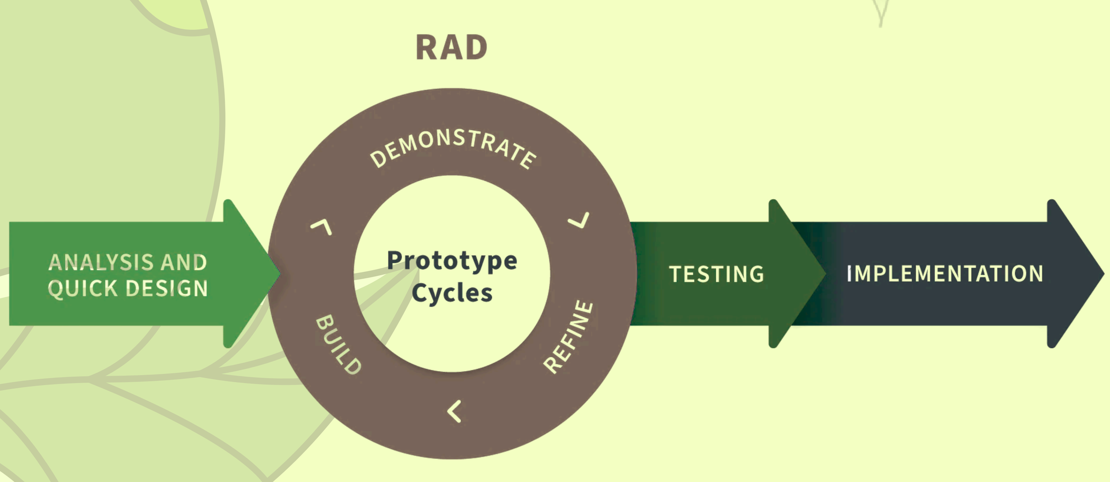
Community Engagement and Literacy

Measures user adoption rates and the effectiveness of the platform in educating residents on recycling protocols.

Data-Driven Governance and Strategy

Tracks the conversion of reporting trends into actionable insights for local infrastructure and policy improvements.

METHODOLOGY



Rapid Application Developement

This study utilized the Rapid Application Development (RAD) methodology to build EcoCheck, prioritizing iterative prototyping and continuous user feedback to ensure a highly usable, adaptive platform for waste management and environmental reporting.

DISCUSSION

The EcoCheck mobile application is designed with a structured architecture and a suite of integrated features that collectively support effective, transparent, and community-driven waste management reporting. Its design emphasizes usability, reliability, and performance, ensuring accessibility for residents and efficient data management for local officials. Each core feature—from waste incident reporting and geotagged image uploads to report tracking and educational content on waste segregation—directly addresses the practical challenges of monitoring illegal dumping and promoting responsible disposal practices. Guided by the ISO/IEC 25010 software quality model, this section evaluates the system's performance across key characteristics such as functional suitability, security, and maintainability. By presenting both the system framework and its high-quality performance ratings, this summary highlights how EcoCheck fosters active citizen participation, encourages environmental accountability, and provides local authorities with the actionable data necessary for long-term sustainability.

RESULTS

- Enhanced accountability and transparency in waste management
- Geotagged reporting and real-time data empower authorities with actionable insights
- Very high usability, accessibility, and functional suitability
- System adheres to ISO/IEC 25010 international quality standards
- Platform fosters citizen participation, promotes responsible behavior, and supports long-term sustainability